

# Simulation and Analysis of 2D EKF Slam with Known and Unknown Correspondences

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**Abstract**—This project is to implement 2-D extended Kalman filter Simultaneous Localization and Mapping (EKF-SLAM) and simulate the algorithm in the environment of MATLAB. The implementation would include both of the situations of known landmarks correspondences and unknown correspondences.

**Index Terms**—SLAM, EKF, Unknown Correspondence

## I. INTRODUCTION

Autonomous vehicles exploring unknown environments rely on Simultaneous Localization and Mapping (SLAM) to navigate effectively. SLAM incrementally estimates both a robot's pose and the positions of surrounding landmarks, even when neither the robot's initial location nor the environment is known [3]. It achieves this through a recursive cycle of predicting the robot's state via onboard sensors or control inputs and then correcting that state using observation data. Although prediction is relatively straightforward, the correction step becomes more complex when landmark correspondences are uncertain, requiring data association to determine whether an observed landmark is new or previously encountered.

Three main categories of SLAM solutions exist: Kalman filter-based, particle filter-based, and graph-based approaches [1], [3], [5], [6]. The probabilistic formulation of SLAM was first established by Smith et al. [5], who showed that landmark estimates become highly correlated through the robot pose uncertainty. Early practical implementations were dominated by Kalman filter-based methods, particularly EKF-SLAM, which became the de facto standard due to their computational efficiency and suitability for small-scale, structured environments. However, reliance on linearization and Gaussian noise assumptions leads to well-known limitations under strong nonlinearities and high uncertainty. To address these issues, particle filter-based methods such as FastSLAM [4] were introduced, enabling more robust handling of non-Gaussian uncertainties and complex data association. More recently, graph-based approaches [6] have emerged as the state-of-the-art solution for large-scale mapping problems, while EKF-SLAM remains a fundamental baseline estimator in controlled settings.

In this project, we focus on EKF-SLAM (Extended Kalman Filter SLAM), a widely studied approach that applies first-order Taylor expansions to linearize both motion and observation models while maintaining a joint Gaussian estimate of robot pose and map features. Our implementation draws

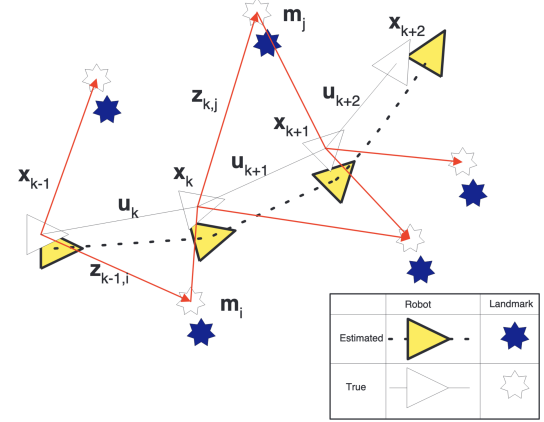


Fig. 1. SLAM Problem : A simultaneous estimate of both robot and landmark location is required. The true locations are never known or measured directly. Observations are made between true robot and landmark locations. Figure courtesy [3]

on methods from *Probabilistic Robotics* [6] and code from the *EL2320 Applied Estimation Labs*. While EKF-SLAM is conceptually elegant, prior work has shown that its performance degrades in the presence of large noise, incorrect data association, and accumulated linearization errors [4], [6]. These limitations motivate an experimental investigation of when EKF-SLAM performs reliably and when it fails.

The goal of this project is therefore to study the behavior of EKF-SLAM under varying noise conditions and data association assumptions. In particular, we compare known and unknown landmark correspondence scenarios and evaluate how estimation accuracy and consistency are affected.

The main contributions of this work include:

- Implementing a 2-D EKF-SLAM algorithm in MATLAB and testing it with map and sensor data from the *EL2320 Applied Estimation* lab files.
- Experimentally evaluating the sensitivity of EKF-SLAM to noise and data association errors.
- Resolving data association related implementation issues in EKF-SLAM based on the pseudocode in *Probabilistic Robotics* [6].

## II. BACKGROUND

### A. SLAM

The SLAM problem has been studied extensively for over two decades and forms a cornerstone of modern mobile

robotics. Its importance arises from the need to operate reliably in environments where prior maps are unavailable or unreliable. Early probabilistic formulations established SLAM as a joint estimation problem, laying the foundation for modern filtering and optimization-based approaches [3], [5].

In probabilistic terms, the goal of Simultaneous Localization and Mapping (SLAM) is to estimate the joint posterior distribution of the robot poses and the positions of surrounding landmarks at time step  $k$ :

$$P(\mathbf{x}_k, \mathbf{m} \mid \mathbf{Z}_{0:k}, \mathbf{U}_{0:k}, \mathbf{x}_0), \quad (1)$$

where  $\mathbf{x}_k$  is the robot's pose at time  $k$ ,  $\mathbf{m}$  denotes the map (i.e., landmark positions),  $\mathbf{Z}_{0:k}$  is the set of all observations up to  $k$ ,  $\mathbf{U}_{0:k}$  is the set of control inputs, and  $\mathbf{x}_0$  is the initial pose. Computing this distribution typically involves a Bayes filter [3], which uses both a motion model and an observation model:

a) *Motion Model.*:

$$P(\mathbf{x}_k \mid \mathbf{x}_{k-1}, \mathbf{u}_k), \quad (2)$$

describes the likelihood of the robot's new pose  $\mathbf{x}_k$  given its previous pose  $\mathbf{x}_{k-1}$  and the control input  $\mathbf{u}_k$ .

b) *Observation Model.*:

$$P(\mathbf{z}_k \mid \mathbf{x}_k, \mathbf{m}), \quad (3)$$

encodes the probability of observing  $\mathbf{z}_k$  from the robot's current pose  $\mathbf{x}_k$  in the presence of known landmarks  $\mathbf{m}$ .

SLAM is then carried out as a recursive Bayes filtering process composed of:

- Prediction:

$$\begin{aligned} & P(\mathbf{x}_k, \mathbf{m} \mid \mathbf{Z}_{0:k-1}, \mathbf{U}_{0:k}, \mathbf{x}_0) \\ &= \int P(\mathbf{x}_k \mid \mathbf{x}_{k-1}, \mathbf{u}_k) \\ & \quad \times P(\mathbf{x}_{k-1}, \mathbf{m} \mid \mathbf{Z}_{0:k-1}, \mathbf{U}_{0:k-1}, \mathbf{x}_0) d\mathbf{x}_{k-1} \end{aligned} \quad (4)$$

- Correction:

$$\begin{aligned} & P(\mathbf{x}_k, \mathbf{m} \mid \mathbf{Z}_{0:k}, \mathbf{U}_{0:k}, \mathbf{x}_0) \\ &= \frac{P(\mathbf{z}_k \mid \mathbf{x}_k, \mathbf{m}) P(\mathbf{x}_k, \mathbf{m} \mid \mathbf{Z}_{0:k-1}, \mathbf{U}_{0:k}, \mathbf{x}_0)}{P(\mathbf{z}_k \mid \mathbf{Z}_{0:k-1}, \mathbf{U}_{0:k})} \end{aligned} \quad (5)$$

where the first step (prediction) propagates the belief over the robot's pose and map based on the motion model, and the second step (correction) incorporates the new observation  $\mathbf{z}_k$  from the observation model.

## B. EKF SLAM

In robotics, EKF-SLAM represents a class of algorithms that applies the Extended Kalman Filter (EKF) to the Simultaneous Localization and Mapping problem. Historically, this approach served as the *de facto* method for SLAM in the 1990s and early 2000s, establishing the foundation for modern robotic navigation [3]. The central contribution of the EKF-SLAM framework is its ability to maintain a fully correlated joint probabilistic estimate over the robot's pose and the locations of observed landmarks.

1) *Mathematical Formulation.*: In the EKF-SLAM framework, the robot's belief state is represented by a mean state vector  $\mu_k$  and a covariance matrix  $\Sigma_k$  at time step  $k$ . In a 2-D planar setting, the state vector is a concatenated column vector containing the robot's pose  $\mathbf{X}_k = [x, y, \theta]^T$  and the coordinates of all  $N$  landmarks  $\mathbf{M}_k = [m_{1,x}, m_{1,y}, \dots, m_{N,x}, m_{N,y}]^T$ :

$$\mu_k = [x \ y \ \theta \ m_{1,x} \ m_{1,y} \ \dots \ m_{N,x} \ m_{N,y}]^T.$$

For convenience, this is often abbreviated as  $\mu_k = [\mathbf{X}_k^T \ \mathbf{M}_k^T]^T$ . The uncertainty of this estimate is maintained in a block-form covariance matrix  $\Sigma_k$ , which captures the correlations between the robot and the landmarks, as well as between different landmarks:

$$\Sigma_k = \begin{pmatrix} \Sigma_{\mathbf{X}\mathbf{X}} & \Sigma_{\mathbf{X}\mathbf{M}} \\ \Sigma_{\mathbf{M}\mathbf{X}} & \Sigma_{\mathbf{M}\mathbf{M}} \end{pmatrix}.$$

Here,  $\Sigma_{\mathbf{X}\mathbf{X}}$  represents the robot pose covariance,  $\Sigma_{\mathbf{M}\mathbf{M}}$  is the map covariance, and  $\Sigma_{\mathbf{X}\mathbf{M}}$  captures the cross-correlations that are essential for the convergence properties of EKF-SLAM [?].

The probabilistic evolution of the state is governed by the standard nonlinear motion and observation models. In the EKF approach, the original probabilistic models are replaced by deterministic functions with additive Gaussian noise:

$$P(\mathbf{x}_k \mid \mathbf{x}_{k-1}, \mathbf{u}_k) \iff \mathbf{x}_k = \mathbf{g}(\mathbf{x}_{k-1}, \mathbf{u}_k) + \mathbf{w}_k, \quad (6)$$

$$P(\mathbf{z}_k \mid \mathbf{x}_k, \mathbf{m}) \iff \mathbf{z}_k = \mathbf{h}(\mathbf{x}_k, \mathbf{m}) + \mathbf{v}_k, \quad (7)$$

where  $\mathbf{g}(\cdot)$  describes the vehicle kinematics,  $\mathbf{h}(\cdot)$  describes the measurement generation process, and  $\mathbf{w}_k$ ,  $\mathbf{v}_k$  are zero-mean, uncorrelated Gaussian noise terms with covariances  $\mathbf{Q}_k$  and  $\mathbf{R}_k$ , respectively.

2) *Initialization and Algorithm.*: The filter is initialized with the robot at the origin with zero uncertainty. Because no landmarks have been observed at  $k = 0$ , the map covariance is initialized with infinite uncertainty. Practically, this is represented as:

$$\mu_0 = [0 \ 0 \ 0 \ \dots \ 0]^T, \quad \Sigma_0 = \begin{pmatrix} 0 & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \infty & \dots & 0 \\ 0 & 0 & 0 & 0 & \dots & \infty \end{pmatrix}.$$

In the case of unknown data association, the initial number of landmarks is  $N_0 = 0$ , and the state vector grows dynamically as new features are incorporated.

The recursive estimation process alternates between a *prediction* phase and a *correction* phase, as outlined in Algorithms 1 and 2 (adapted from [6]).

- 1) **Prediction:** The robot's motion control inputs  $\mathbf{u}_k$  are processed using the motion model  $\mathbf{g}(\cdot)$  to propagate the state estimate and covariance. A dimensionality matrix  $F_x$  is often employed to map the low-dimensional vehicle state into the high-dimensional full state vector during Jacobian calculations.

- 2) **Correction:** When measurements  $z_k$  are available, the filter updates the state using the Kalman gain, reducing uncertainty in both the robot pose and the map.

Although these algorithms largely follow the textbook formulations, minor modifications are often made for implementation convenience, such as specific memory management steps in MATLAB.

3) **Data Association:** Data association—determining which landmark in the map corresponds to a current sensor reading—is critical for EKF-SLAM performance.

**Known Correspondences:** When feature identities are known, the estimator strictly updates existing landmarks or initializes new ones based on a state lookup. The position of a new landmark is initialized using the inverse of the measurement model relative to the current robot pose.

**Unknown Correspondences:** In realistic settings, the algorithm must solve the association ambiguity. Typically, a Maximum Likelihood (ML) approach is used. The Mahalanobis distance is calculated between the measurement and the predicted observations of all existing landmarks. If the minimum distance falls below a statistical threshold  $\alpha$ , the measurement is associated with that landmark. If the distance exceeds the threshold for all landmarks, the measurement is treated as a new feature and added to the state vector.

Incorrect association decisions can be catastrophic: they introduce spurious landmarks and corrupt the joint covariance structure. Because the EKF maintains a single Gaussian hypothesis, these errors are difficult to recover from and often lead to divergence [1], [2].

4) **Limitations:** Despite its theoretical elegance, EKF-SLAM suffers from fundamental limitations rooted in its core assumptions.

First, the algorithm relies on the first-order Taylor series expansion to linearize the nonlinear functions  $g(\cdot)$  and  $h(\cdot)$ . When the uncertainty in the state estimate is large, this linearization becomes invalid, causing the estimated covariance to become overly optimistic (inconsistent) compared to the true error [?], [6].

Second, the EKF inherently assumes Gaussian noise distributions. In real-world environments characterized by clutter and ambiguous data association, the true posterior is often multi-modal and non-Gaussian. The EKF cannot represent these multiple hypotheses; once an incorrect data association is fused into the Gaussian belief, it cannot be easily undone. These weaknesses, particularly the inability to robustly handle non-Gaussian uncertainty and the computational scaling with respect to map size, motivated the development of particle-filter approaches such as FastSLAM [4].

### III. IMPLEMENTATION AND SIMULATION IN MATLAB

#### A. Implementation

The main structure of this project is inspired from lab files from EL2320. The algorithm part of the codes is divided into functions, predict and correct for easier understanding and testing.

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#### Algorithm 1 EKF-SLAM with Known Correspondences

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**Require:** Posterior at time  $t - 1$ :  $(\mu_{t-1}, \Sigma_{t-1})$

Control signals at time  $t$ :  $u_t$

Observations at time  $t$ :  $z_t = \{z_t^1, z_t^2, \dots\}$

Known landmark correspondences at time  $t$ :  $c_t$

**Ensure:** Posterior at time  $t$ :  $(\mu_t, \Sigma_t)$

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0: **Predict Step**

0:  $(\bar{\mu}_t, \bar{\Sigma}_t) \leftarrow \text{predict}(\mu_{t-1}, \Sigma_{t-1}, u_t)$  {Apply motion model and linearize to obtain predicted mean and covariance.}

0: **Correction Step**

0: **for each** observed feature  $z_t^i$  in  $z_t$  **do**

0:  $j \leftarrow c_t^i$  {Landmark index from correspondence.}

0: **if** landmark  $j$  is new **then**

0: initialize landmark  $j$  in state {Use  $z_t^i$  and current pose to add a new landmark.}

0: **end if**

0:  $(\hat{z}_t^i, H_t^i) \leftarrow \text{expectedObservation}(\bar{\mu}_t, j)$  {Compute predicted measurement and Jacobian.}

0:  $K_t^i \leftarrow \text{kalmGain}(\bar{\Sigma}_t, H_t^i)$  {Compute Kalman gain using  $\bar{\Sigma}_t$  and  $H_t^i$ .}

0:  $(\mu_t, \Sigma_t) \leftarrow \text{update}(\bar{\mu}_t, \bar{\Sigma}_t, K_t^i, z_t^i, \hat{z}_t^i, H_t^i)$  {Update mean and covariance.}

0:  $(\bar{\mu}_t, \bar{\Sigma}_t) \leftarrow (\mu_t, \Sigma_t)$  {Set up for next feature's correction.}

0: **end for**

0: **return**  $(\mu_t, \Sigma_t)$   
=0

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The motion model is simplified in this implementation, hence the new prediction step would be,

$$\bar{\mu}_t = \mu_{t-1} + F_x^T \begin{pmatrix} v_t \Delta t \cos \mu_{t-1, \theta} \\ v_t \Delta t \sin \mu_{t-1, \theta} \\ \omega_t \Delta t \end{pmatrix} \quad (8)$$

$$G_t = I + F_x^T \begin{pmatrix} 0 & 0 & -v_t \Delta t \sin \mu_{t-1, \theta} \\ 0 & 0 & v_t \Delta t \cos \mu_{t-1, \theta} \\ 0 & 0 & 0 \end{pmatrix} F_x$$

$$= I + F_x^T \begin{pmatrix} 0 & 0 & -\bar{\mu}_{t,y} \\ 0 & 0 & \bar{\mu}_{t,x} \\ 0 & 0 & 0 \end{pmatrix} F_x \quad (9)$$

1) **Known Data Association:** One main challenge in SLAM is deciding if a landmark is new or already in the map. In the known data association scenario, this project uses a matrix called `landmark_obs_flag` to track each landmark's observation status.

2) **Unknown Data Association:** When implementing EKF-SLAM with unknown data association, a key issue arises in the **for** loop at line 8 of Algorithm 2. Specifically, when  $k = N_t + 1$ , the term  $2N_t - 2k$  becomes  $-2$ . This term defines the width of a zero block in the matrix  $F_{x,k}$ , which must not be negative. The matrix size is set this way to match  $\bar{\Sigma}_t$  from the predict step, which is of size  $(2N_t + 3) \times (2N_t + 3)$ .

To address this, the implementation expands  $\bar{\Sigma}_t$  before the **for** loop:

$$\bar{\Sigma}_t = \begin{pmatrix} \bar{\Sigma}_t & \mathbf{0}^T \\ \mathbf{0} & \infty \end{pmatrix},$$

where

$$\mathbf{0} = \begin{pmatrix} 0 & \dots & 0 \\ \underbrace{0 & \dots & 0}_{2N_t+3} \end{pmatrix}, \quad \infty = \begin{pmatrix} \infty & 0 \\ 0 & \infty \end{pmatrix}.$$

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**Algorithm 2** EKF-SLAM with Unknown Correspondences

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**Require:** Posterior at time  $t-1$ :  $(\mu_{t-1}, \Sigma_{t-1})$   
Control input at time  $t$ :  $u_t$   
Observations at time  $t$ :  $z_t = \{z_t^1, z_t^2, \dots\}$   
Number of landmarks from previous step:  $N_{t-1}$   
**Ensure:** Posterior at time  $t$ :  $(\mu_t, \Sigma_t)$   
Updated number of landmarks:  $N_t$

0: **Initialize:**  
0:  $N_t \leftarrow N_{t-1}$  {Start with the same number of landmarks as before.}

0: **Predict Step**  
0:  $(\bar{\mu}_t, \bar{\Sigma}_t) \leftarrow \text{predict}(\mu_{t-1}, \Sigma_{t-1}, u_t)$  {Apply motion model and linearization (EKF prediction).}

0: **Correction / Data Association**  
0: **for each** observed feature  $z_t^i \in z_t$  **do**  
(a) *Consider Each Possible Landmark (Including "New" Landmark)*  
0: **for**  $k = 1 \rightarrow N_t + 1$  **do**  
0: Compute the *innovation* or difference  $\delta_k^i = z_t^i - \hat{z}_t^i(k)$  { $\hat{z}_t^i(k)$  is the expected measurement for landmark  $k$ .}  
0: Evaluate the *likelihood* or *cost*  $\pi_k^i$  of matching  $z_t^i$  to landmark  $k$  {e.g., Mahalanobis distance, gating test, etc.}  
0: **end for**  
(b) *Select the Best Match or Initialize a New Landmark*  
0:  $j(i) \leftarrow \arg \min_k \pi_k^i$  {Choose the landmark  $j(i)$  that yields minimum cost.}  
0: **if**  $\pi_{j(i)}^i > \text{threshold}$  **then**  
0:  $j(i) \leftarrow N_t + 1$  {Treat  $z_t^i$  as a new landmark if best cost is too large.}  
0:  $N_t \leftarrow N_t + 1$  {Increase landmark count by one.}  
0: **end if**  
(c) *EKF Update for the Chosen Landmark  $j(i)$*   
0:  $(\mu_t, \Sigma_t) \leftarrow \text{update}(\bar{\mu}_t, \bar{\Sigma}_t, z_t^i, j(i))$  {Apply standard EKF correction (using the chosen landmark).}  
0:  $(\bar{\mu}_t, \bar{\Sigma}_t) \leftarrow (\mu_t, \Sigma_t)$   
0: **end for**

0: **return**  $(\mu_t, \Sigma_t)$  and  $N_t$   
=0

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After this operation,  $\bar{\Sigma}_t$  becomes  $[2(N_t + 1) + 3] \times [2(N_t + 1) + 3]$ . If the observed landmark is not new,  $\bar{\Sigma}_t$  is then trimmed back to its original size:

$$\bar{\Sigma}_t \leftarrow \bar{\Sigma}_t(1 : (2N_t + 3), 1 : (2N_t + 3)).$$

Additionally, the number of zeros in  $F_{x,k}$  becomes  $2(N_t + 1) - 2k$ , ensuring it is never negative.

### B. Results and Analysis

The EKF-SLAM simulations were recorded as .gifs, which are located in the GIFs folder.

#### 1) Known Data Association:

a) *map\_o3.txt + so\_o3\_ie.txt*: According to the description in *Lab of EL2320 Applied Estimation*, the laser scanner has an accuracy of approximately 1 cm and 1 degree, while the odometry data contains unmodeled noise on the order of 1 cm and 1 degree per timestep. Consequently, the motion disturbance and observation noise matrices are set as:

$$R = \begin{pmatrix} 0.01^2 & 0 & 0 \\ 0 & 0.01^2 & 0 \\ 0 & 0 & (\frac{2\pi}{360})^2 \end{pmatrix}, \quad Q = \begin{pmatrix} 0.01^2 & 0 \\ 0 & (\frac{2\pi}{360})^2 \end{pmatrix}.$$

Figures 2, 3, and 4 (along with the accompanying video *map\_o3+Video.gif*) illustrate the simulation results, where the estimated robot pose and landmark positions closely match the ground truth. While the small errors and covariance values suggest that the EKF-SLAM algorithm performs well on this dataset, a more critical observation in our analysis

is the *filter consistency*. By comparing the estimation error (Fig. 3) with the state variance (Fig. 4), we observe that the error for the  $x$ -position (peak  $\approx 0.01\text{m}$ ) remains well within the  $3\sigma$  uncertainty envelope ( $\sqrt{\Sigma_{1,1}} \approx 0.07\text{m}$ ). This indicates that the filter is not only accurate but also maintains a consistent internal representation of its own uncertainty, effectively validating the choice of noise parameters  $R$  and  $Q$ .

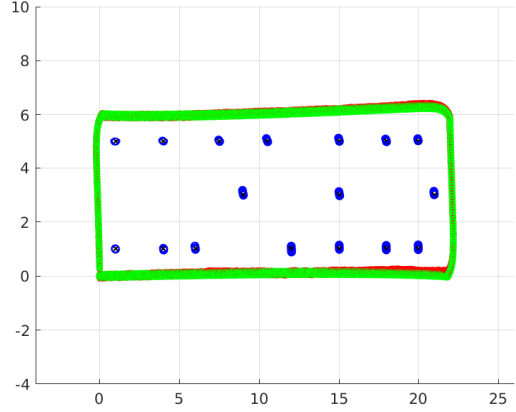


Fig. 2. Trace of Robot Pose and Map Location

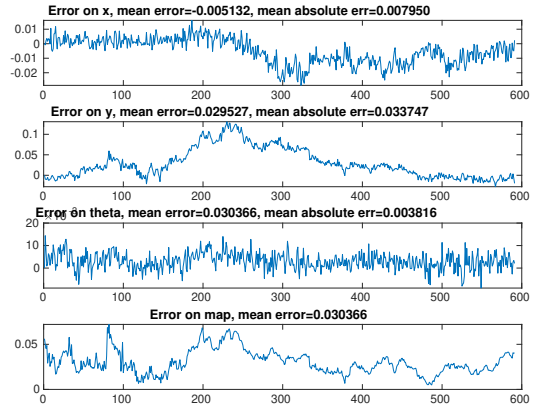


Fig. 3. Error of Robot Pose and Average Map Location

b) *map\_pent\_big\_40.txt + so\_pb\_40\_no.txt*: In this dataset, measurement noise is modeled with a standard deviation of 0.1 (meters, radians), while process noise is assigned a standard deviation of 1 (meters, meters, radians). The map forms a highly symmetric circular arrangement of landmarks, and the disturbance and noise are:

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 0.1 & 0 \\ 0 & 0.1 \end{pmatrix}.$$

Figures 5, 6, and 7 illustrate a critical failure case where the EKF-SLAM algorithm performs poorly, producing incorrect

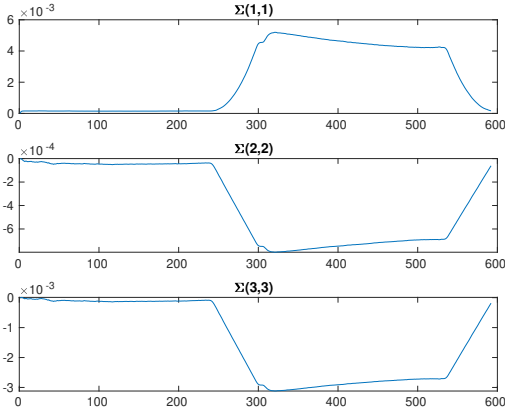


Fig. 4. Variance of Robot Pose

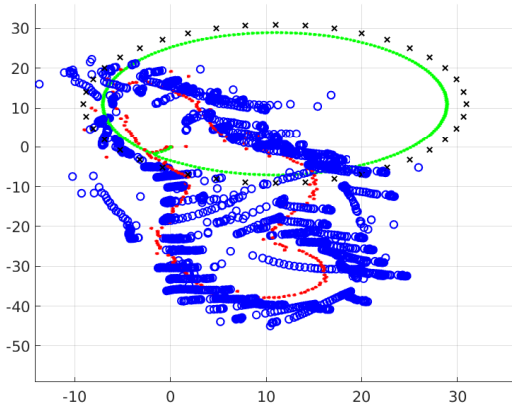


Fig. 5. Trace of Robot Pose and Map Location

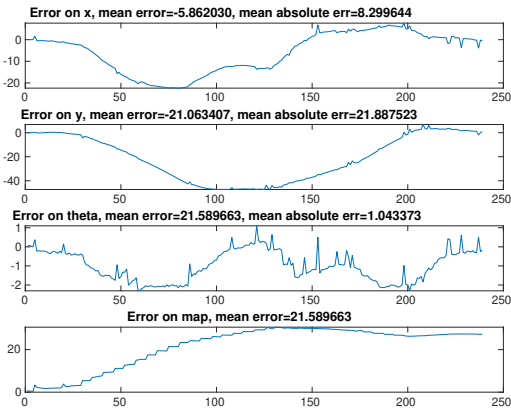


Fig. 6. Error of Robot Pose and Average Map Location

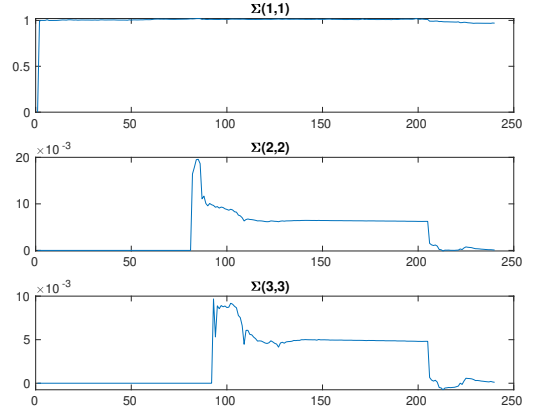


Fig. 7. Error of Robot Pose and Average Map Location

estimates of both robot and landmark locations. While the estimation error grows significantly (Fig. 6), the variance  $\Sigma$  (Fig. 7) remains dangerously small. This is a classic example of *filter inconsistency* or *overconfidence*, a fundamental theoretical risk of the EKF discussed by Bailey et al. [5]. This failure is primarily driven by two factors:

- **Strong nonlinearities and linearization error:** Because the EKF linearizes the non-linear motion model via a first-order Taylor expansion around the current estimate, large process noise causes the robot pose to deviate from the true state. Consequently, the Jacobians  $G_t$  in Eq. 9 are evaluated at an incorrect linearization point. As noted by Thrun [6], this leads to the filter "trusting" incorrect measurements too heavily, causing the covariance to shrink even as the actual error explodes.
- **Outlier sensitivity in data association:** While outliers can be detected via Mahalanobis distance thresholds in pure localization, they are catastrophic in SLAM. An observation with large error may be incorrectly treated as a new landmark, corrupting the joint covariance structure and making subsequent outlier detection more difficult.

Our results confirm that once the initial uncertainty is high, the EKF ignores new sensor data in favor of its own corrupted internal state, leading to the complete divergence of the map seen in Figure 5.

2) *Unknown Data Association:* The same two datasets are tested under unknown data association, with identical motion disturbance and observation noise settings.

a) *map\_o3.txt + so\_o3\_ie.txt:* Figures 8, 9, and 10 show the results. Compared to the known data association case, the unknown scenario exhibits larger errors, possibly due to landmarks that are not correctly associated with measurements. These unassociated landmarks still affect the overall state, so if their positions deviate from reality, they can degrade the overall accuracy.

b) *map\_pent\_big\_40.txt + so\_pb\_40\_no.txt:* Figures 11, 12, and 13 illustrate the results. Although the SLAM process appears initially reasonable (more clearly

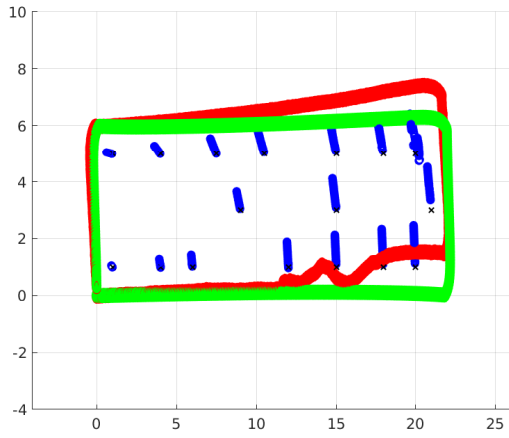


Fig. 8. Trace of Robot Pose and Map Location

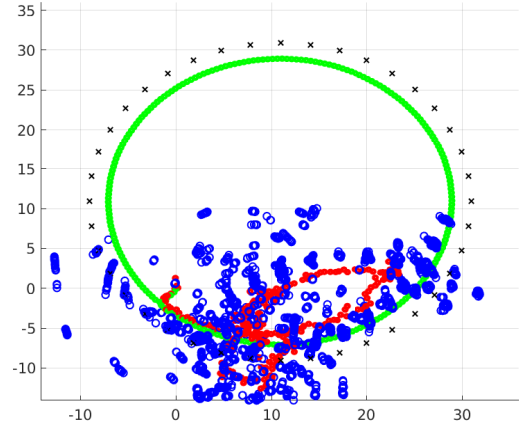


Fig. 11. Trace of Robot Pose and Map Location

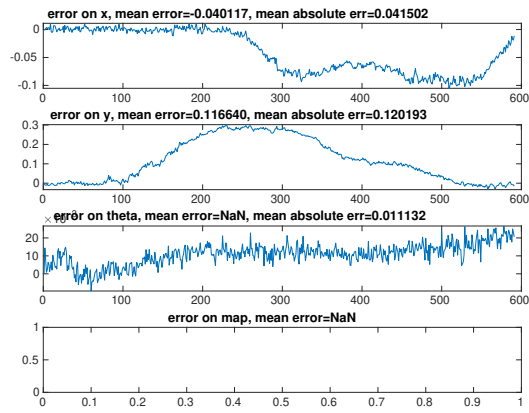


Fig. 9. Error of Robot Pose and Average Map Location

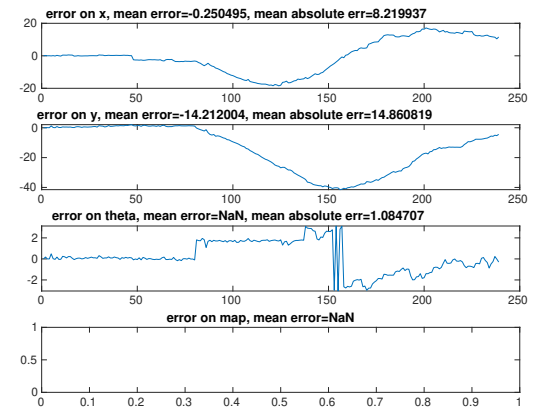


Fig. 12. Error of Robot Pose and Average Map Location

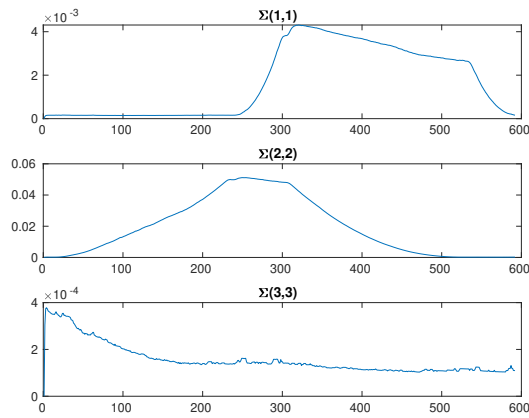


Fig. 10. Variance of Robot Pose

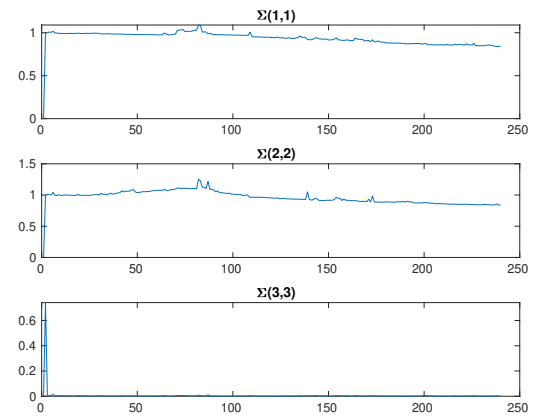


Fig. 13. Error of Robot Pose and Average Map Location

shown in the accompanying video), a sudden change in robot orientation occurs, likely caused by an outlier observation. This leads to significant errors in subsequent pose and landmark estimates.

In the Unknown Correspondence scenario (Figures 9 and 12), the error metrics eventually result in 'NaN' values, serving as a practical demonstration of the fundamental limitations of single-hypothesis filters. Our analysis suggests this divergence is due to the catastrophic failure of the Mahalanobis distance gate; a single incorrect data association incorporates a high-innovation measurement into the state  $\mu$ , which can cause the covariance matrix  $\Sigma$  to lose its positive-definiteness during the update step (Eq. 5). This numerical instability is a known limitation of the EKF's single-hypothesis nature [5], as it cannot represent the multi-modal distributions that arise when a measurement could correspond to multiple different landmarks—a common occurrence in high-clutter environments. This fragility is a primary motivation cited by Montemerlo et al. [4] for the development of multi-hypothesis approaches like FastSLAM, which utilize particle filters to track multiple data association paths simultaneously to avoid the estimator collapse observed in our experiments.

#### IV. CONCLUSION

In this project, I successfully implemented and analyzed a 2-D EKF-SLAM framework in MATLAB, utilizing laboratory datasets and the algorithmic foundations in [6]. A primary implementation contribution involved resolving numerical ambiguities in the dynamic expansion of the state vector and covariance matrix during unknown data association—specifically ensuring matrix consistency when initializing new landmarks.

The experimental results demonstrate that while EKF-SLAM is a computationally efficient baseline that provides accurate, consistent estimates in structured environments with precise measurements (as seen in Figs 2,3,4), it is fundamentally sensitive to noise and data association accuracy. The divergence observed in our "Unknown" correspondence simulations (Figs 11,12) highlights the "overconfidence" problem inherent in the EKF. Because the filter linearizes around a single Gaussian mean, a single incorrect association corrupts the linearization point for the Jacobians, causing the filter to trust its own erroneous state over new sensor data.

These findings reinforce the well-known theoretical limitations of first-order linearization and single-hypothesis tracking. Historically, the failure modes observed in this project—specifically the inability to recover from incorrect associations and the resulting numerical instability (NaN values)—motivated the shift in the state-of-the-art toward multi-hypothesis trackers such as FastSLAM [4] and modern Factor Graph optimizations [3]. In conclusion, this report clarifies that while EKF-SLAM remains a powerful tool for low-uncertainty applications, robust autonomous navigation in unstructured environments requires the more advanced filtering and optimization strategies that have succeeded it.

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